Sturm-Liouville Theory: An Example

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Methods in Applied Mathematics

December 14, 2018

Outline



Background

Theorems

Example: Heat Flow without Source

Example: Heat Flow with Source

Example: Computational Example

What is Sturm-Liouville Theory?

- Theory of a real second-order linear differential equation of the form

$$\frac{d}{dx}\left(p(x)\frac{d\phi}{dx}\right) + q(x)\phi(x) + \lambda\sigma(x)\phi(x) = 0.$$

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- λ corresponds to an *eigenfunction*.

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- Regular Sturm-Liouville Eigenvalue problem

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- *Boundary Conditions*

$$\beta_1\phi(a) + \beta_2\frac{d\phi}{dx}(a) = 0$$

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- $p > 0, \sigma > 0$ and q are real valued and continuous everywhere including endpoints

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3. Each λ_n corresponds to a unique (up to a multiple) of $\phi_n(x)$, an eigenfunction. This eigenfunction has $n-1$ zeros in (a, b).
4. The set $\{\phi_n(x)\}$ is complete and

$$f(x) \sim \sum_{n=1}^{\infty} a_n \phi_n(x)$$

for f piecewise smooth.

Main Theorems of Sturm-Liouville

5 The set $\{\phi_n(x)\}$ is orthogonal with respect to $\sigma(x)$, that is

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6 Eigenvalues can be related to eigenfunctions by *Rayleigh Quotient*:

$$\lambda = \frac{-p\phi\ \dfrac{d\phi}{dx}\Big|_a^b + \displaystyle\int_a^b p\left(\frac{d\phi}{dx}\right)^2 - q\phi^2\ dx}{\displaystyle\int_a^b \phi^2\sigma\ dx}$$

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PDE and BCs are linear and homogeneous (ignore ICs for now). Solution is of form

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Then divide through to get

$$\frac{h'(t)}{h(t)} = \frac{\frac{\partial}{\partial x}\big(K_0\phi'(x)\big)}{c\rho\phi(x)} = -\lambda$$

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$$\frac{d}{dx}\big(K_0\phi'(x)\big) + \lambda c\rho\phi(x) = 0 \tag{2}$$

with BCs: $\phi(0) = \phi'(L) = 0$.

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We can solve (1) easily:

$$\frac{dh}{h} = -\lambda\, dt \Longrightarrow h(t) = ce^{-\lambda t} \tag{3}$$

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- By superposition the linear combination of u_k will also be a solution, that is

$$u(x,t) = \sum_{k=1}^{\infty} a_k \phi_k(x) e^{-\lambda_k t}. \tag{4}$$

Initial Conditions and a_k

$$u(x,t) = \sum_{k=1}^{\infty} a_k \phi_k(x) e^{-\lambda_k t}$$

solves our PDE, let $t = 0$ to get

$$f(x) = u(x,0) = \sum_{k=1}^{\infty} a_k \phi_k(x) \tag{5}$$

Recall Theorem 4 that for f smooth we can express as a Fourier series.

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By orthogonality

$$\int_0^L \phi_n(x)\phi_m(x)c(x)\rho(x)\, dx = 0 \quad n \neq m. \tag{6}$$

and $\sigma(x) = c(x)\rho(x)$ is our weight function.

Initial Conditions and a_k

To get our Fourier coefficients multiply (5) by the weight function and eigenfunction, that is

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by orthogonality only the term when $k = j$ survives:

$$\int_0^L f(x)\phi_j(x)c(x)\rho(x)\ dx = a_j \int_0^L \phi_j(x)^2 c(x)\rho(x)\ dx$$

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To get the eigenvalues recall Theorem 6 and let $p = K_0(x)$, $q(x) = 0$, and $\sigma(x) = c(x)\rho(x)$ to get

$$\lambda_n = \frac{\int_0^L K_0(x)\left(\frac{d\phi_n}{dx}\right)^2\ dx}{\int_0^L \phi_n(x)^2 c(x)\rho(x)\ dx}. \tag{8}$$

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- Therefore we modify λ_k to get

$$\lambda_k = \frac{\displaystyle\int_0^L K_0(x)\left(\frac{d\phi_k}{dx}\right)^2 - \alpha(x)\phi_n(x)\,dx}{\displaystyle\int_0^L \phi_n(x)^2 c(x)\rho(x)\,dx}$$

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We need to consider 3 cases: $\lambda = 0$, $\lambda < 0$, and $\lambda > 0$.

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- Suppose $\lambda = 0$, then $y'' = 0$. Therefore

$$y(x) = Ax + B$$

with the first boundary condition we get

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- Suppose $\lambda < 0$, then α is a real number since $\alpha = \pm\sqrt{-\lambda}$. Therefore

$$y(x) = Ae^{-\sqrt{-\lambda}x} + Be^{\sqrt{-\lambda}x}$$

with our BCs we get the following system

$$\begin{cases} A(1 - e^{-\sqrt{-\lambda}\pi}) + B(1 - e^{\sqrt{-\lambda}\pi}) = 0 \\ A(-1 + e^{-\sqrt{-\lambda}\pi}) + B(1 - e^{\sqrt{-\lambda}\pi}) = 0 \end{cases}$$

Computational Example

- For the system the only solution is $A = B = 0$ which leads to the trivial solution.
- Suppose $\lambda > 0$ then α is imaginary and we get that $y(x)$ is a linear combination of sine and cosine:

$$y(x) = A\cos\left(\sqrt{|\lambda|}x\right) + B\sin\left(\sqrt{|\lambda|}x\right)$$

using the BCs we get the following system

$$\begin{cases} A\left(1 - \cos\left(\sqrt{|\lambda|}\pi\right)\right) - B\sin\left(\sqrt{|\lambda|}\pi\right) = 0 \\ A\sin\left(\sqrt{|\lambda|}\pi\right) + B\left(1 - \cos\left(\sqrt{|\lambda|}\pi\right)\right) = 0 \end{cases}$$

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We can write the system as a matrix problem

$$\begin{pmatrix} 1-\cos\left(\sqrt{|\lambda|}\pi\right) & -\sin\left(\sqrt{|\lambda|}\pi\right) \\ \sin\left(\sqrt{|\lambda|}\pi\right) & 1-\cos\left(\sqrt{|\lambda|}\pi\right) \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

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$$\left(1-\cos\left(\sqrt{|\lambda|}\pi\right)\right)^2 + \sin^2\left(\sqrt{|\lambda|}\pi\right) = 0$$

Which reduces to

$$2-2\cos\left(\sqrt{|\lambda|}\pi\right) = 0 \Longrightarrow \cos\left(\sqrt{|\lambda|}\pi\right) = 1.$$

This has solutions when $\sqrt{|\lambda|} = 2, 4, 8, \ldots$

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Therefore, $\lambda_n = 4n^2$ for $n = 1, 2, 3, \ldots$, these are our eigenvalues.

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The corresponding eigenfunctions are given by

$$\phi_n(x) = \cos(2nx) \quad \psi_n(x) = \sin(2nx).$$

References I



R. Haberman.
Applied Partial Differential Equations, with Fourier Series and Boundary Value Problems, Fifth Edition.
Pearson, 2013.